



Power Module

Installation and user manual

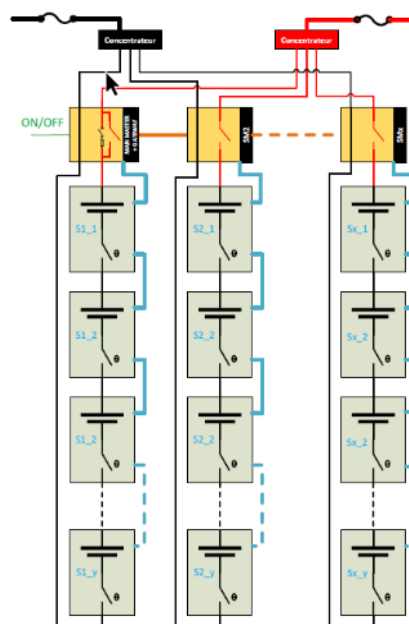
V1.8

Single Module
51,2V 105Ah, 5,37KWh



Large serial/parallel assembly
configuration with SuperMaster

up to 128 modules, 819.2V 688KWh





Index of changes				
Index	Date	P. modified	Description of changes	Author
V0.1	10/12/20	creation		Tech Dpt
V0.2	02/09/21	creation	Wording	Tech Dpt
V0.3	20/10/21	update	CAN Dictionary	Tech Dpt
V0.4	03/11/21	Update	Wording	Tech Dpt
V1.0	16/09/22	Update	Termination and maintenance	Tech Dpt
V1.1	13/04/23	Update	caution parallel assembly	Tech Dpt
V1.2	06/02/24	Update	Comissionning + LED End of Charge	Tech Dpt
V1.3	05/03/24	Update	PowerModule with Precharge	Tech Dpt
V1.4	08/03/24	Update	Status 2 SuperMaster	Tech Dpt
V1.5	12/03/24	Update	Charge and balancing	Tech Dpt
V1.6	22/03/24	Update	CAN decoding explanation	Tech Dpt
V1.7	23/07/2023	Update	Handling CAN Protocol	Tech Dpt
V1.8	20/11/2023	Update	Angle torque and SuperMaster diagram	Tech Dpt



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1 Introduction

The PowerModule System is a modular, intelligent and scalable battery, using a storage technology that offers exceptional performance in terms of available power, energy density and lifetime. The most advanced configurations can be connected to the Internet (cloud) for remote monitoring.

The PowerModule is an advanced Lithium battery for industrial vehicles, medium to heavy duty traction, robotics and applications requiring high capacity and/or high voltage up to 819.2 V nominal. Up to 128 modules can be assembled in series, parallel or both.

For small systems with up to 16 modules, which do not require cloud-based monitoring or precharge system, an external BMS is not required. An additional external BMS, called a Supermaster, allows for cloud-based monitoring, and is required in the case of an assembly with a large number of PowerModules.

The system incorporates intelligent BMSMatrix® technology that makes deployment and maintenance much easier and ensures reliable battery operation for many years. All the complexity inherent in the proper operation and safety of LiFePO₄ batteries is transparent to the user and is included in the electronics and algorithms embedded in the BMSMatrix® technology.

This manual gives the technical specifications and operating ranges of the PowerModule. It also describes the PowerModule assemblies.

Important : for proper commissioning, please refer to the chapter on " **First commissioning and charging** ".

Highlights of the Li-ion PowerModule system:

- BMSMatrix®: Full scalability from 50V to 819VDC and 5kWh to 688KWh
- Extremely resistant stainless steel stackable enclosures
- 96% energy efficiency (Roundtrip efficiency)
- Choice of on-board technology: Lithium Iron Phosphate (LiFePO₄ or LFP) technology that is totally safe and inherently stable, offering a long service life
- Several thousand charge/discharge cycles available
- Deep discharge with no noticeable effect on battery life
- No memory effects
- No energy loss at high discharge levels (extremely low Peukert losses)
- Fast charge time compared to other battery types
- BMS (Battery Management System) embedded in each PowerModule ensuring total control of the battery (cell voltages, SoC, SoH, current measurement, balancing, safety cut-off)
- Communication via CAN 2.0B bus.
- Power cut assured by power contactor.
- Fuse integrated in each PowerModule for a second level of safety
- Management of a pre-charge circuit integrated in the SuperMaster.
- 70% weight saving compared to a lead acid battery
- Reduced size
- Maintenance free, no liquid levels to be done
- No off-gassing



1.1 Use of this manual

This manual is a guide to the installation, use and maintenance of PowerModules batteries. It is essential that every person working on or with a LiFePO₄ battery is familiar with the contents of this manual and that he/she follows the instructions contained herein carefully.

Installation of and work on a Li-ion battery should only be carried out by authorized, qualified and trained personnel, acting in accordance with the locally applicable standards and safety recommendations.

For installations with several modules, it is important to understand the installation of a single module

1.2 Warranty limits

PowerTech Systems warrants that the LiFePO₄ battery has been manufactured in accordance with applicable legal standards and specifications. Operations not in accordance with the instructions in this manual may damage the battery or alter its characteristics. In such cases the warranty becomes null and void.

The warranty is limited to the repair and/or replacement of the product. It does not cover the cost of installation labor or shipping of defective parts.



CAUTION: Damage to the battery due to prolonged and sustained deep discharge is not covered by the warranty.

The period of validity and the conditions of application of the warranty can be found in the general sales conditions.

PowerTech Systems accepts no responsibility for:

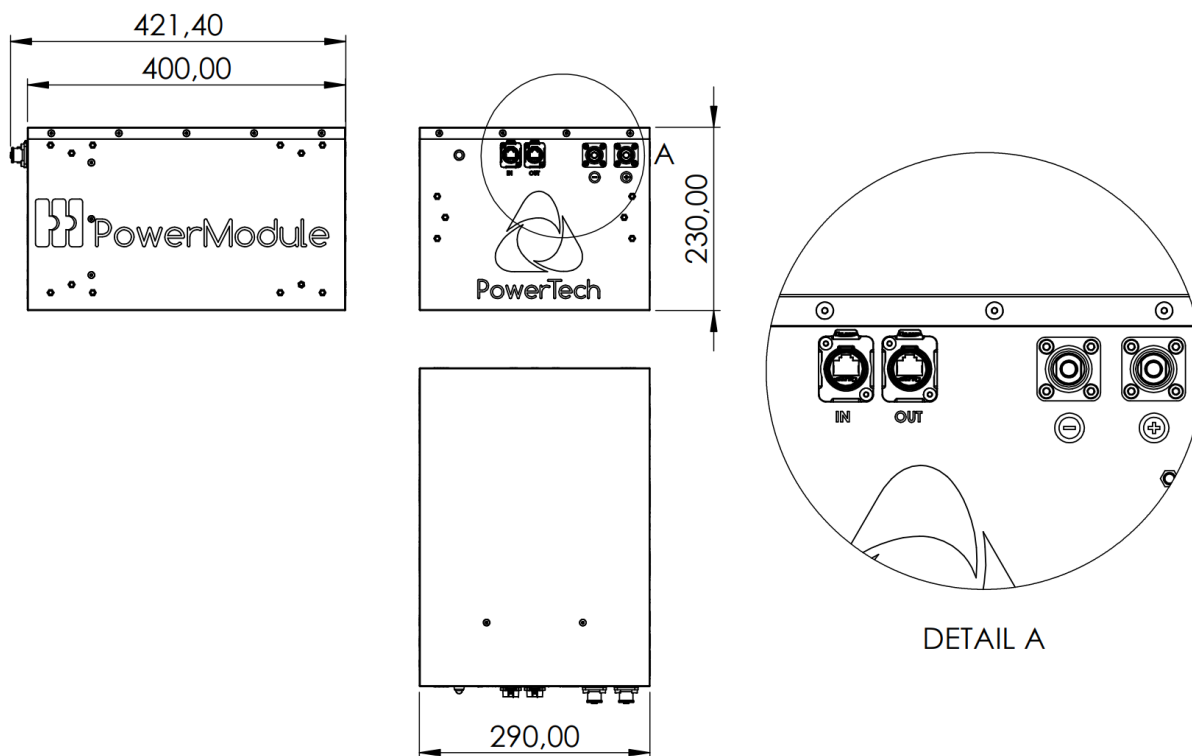
- Consequential damages resulting from the use of the PowerModule product
- Possible errors in this manual and their consequences



2 Characteristics of a single module



2.1 Dimensions



Note: on the first versions of the PowerModule the IN and OUT plugs are reversed



2.2 Technical Specifications

Technology	Lithium Iron Phosphate (LFP - LiFePo4)
Min / Nominal / Max voltage	48.0 V / 51.2 V / 58.4 V
Capacity / Energy stored (at 1C, 25°C)	105Ah/5.376 KWh
Weight (+/- 3 %)	43.5 Kg
Dimensions (l x w x h)	400 x 290 x 230 mm
Operating temperature	from -20°C, to +60°C
Negative Temperature	Integrated heating system
Power connector	Amphenol Powerlok IP67
Specific energy	123.5 Wh/Kg
Energy density	201.5 Wh/l
Continuous discharge current (at 20 °C)	125 A (6.40kW)
Peak discharge current (5 minutes)	200 A (10.24kW)
Peak discharge current (30 seconds)	250 A (12.80kW)
Recommended charging voltage	57.0 V
Maximum charging voltage	58,4 V
Recommended float voltage	54.0 V
Standard charging current	50 A (2.56kW)
Fast charging current	100 A (5.12kW)
Orientation	No orientation sensitive elements
Protection class	IP54 (IP 66 available upon request)
Self-Discharge	~5% per month in storage

2.3 Fuse

Each PowerModule is equipped with a 250 Amp fuse, which should only act in case of a serious problem. To protect the operating circuits from false operation, you must install a smaller fuse upstream of the system. It is recommended to put a fuse on each pole of the battery.

2.4 Internal Heating

The internal heating consists of a resistor powered by the battery. It is controlled by BUS-CAN commands for switching on and off as described in the BUS-CAN protocol chapter. A command allows the automatic heating on load if the temperature is low.

2.5 Estimated number of cycles

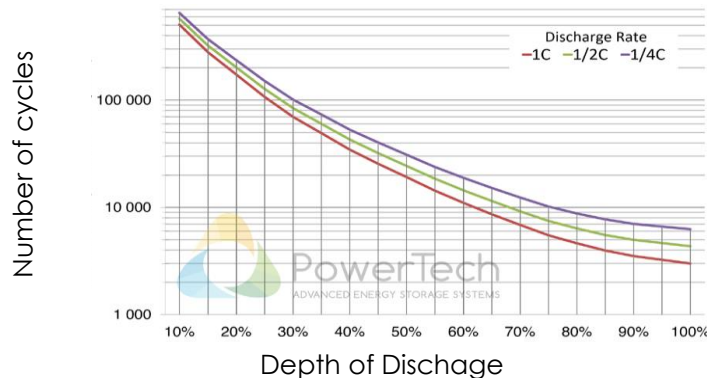
Test conditions: charge at constant C/4, discharge at constant C/2, ambient temperature of 25°C.

80% residual energy remaining at the end of the cycles.

at 70% DOD ~6000
at 80% DOD ~4000
at 90% of DOD ~3400
at 100% DOD ~3000

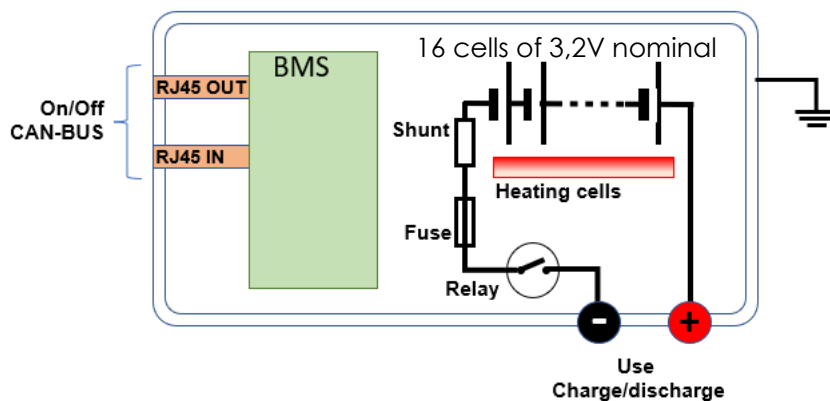


Number of cycles depending on the depth of discharge (DOD)



The number of cycles also varies with the cycling temperature. Up to 35°C the SOH is not too much affected. At 45°C, the loss of SOH is noticeable.

2.6 PowerModule internal diagram



2.7 BMS System

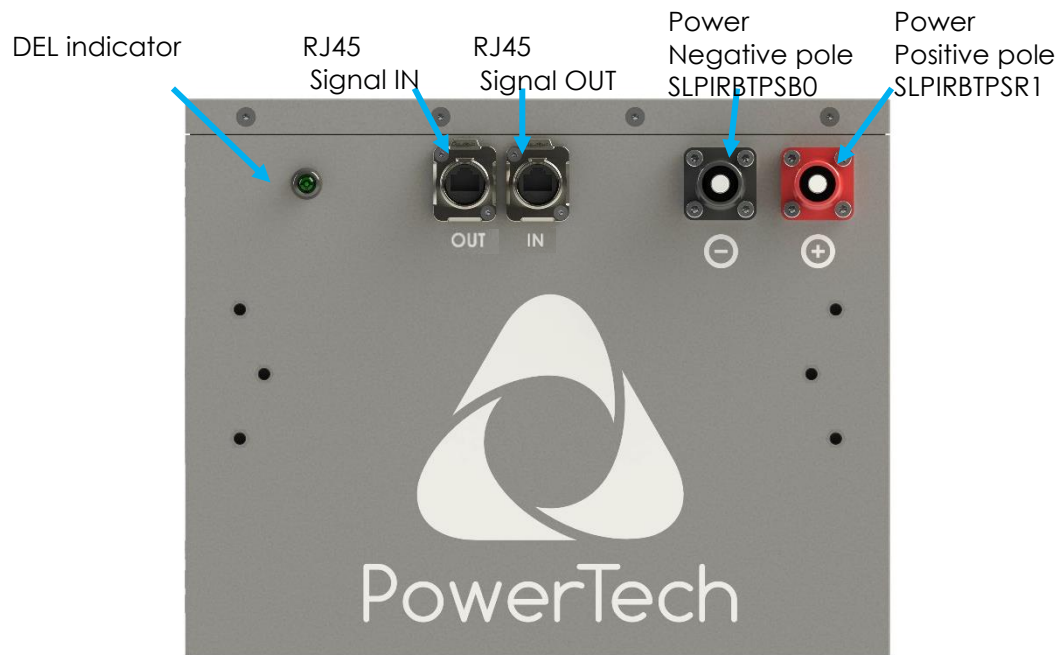
Features :

- Balancing of the 16 cells composing the PowerModule
- Monitoring of the pack and cell voltage
- Real time monitoring of the delivered current
- Precise calculation of the SoC (State of Charge) by precision shunt and Hall effect sensors
- Measurement of the SoH (State of Health) by means of algorithms.
- Monitoring of the temperature of the electronics and the cells
- Security via power contactor
- Digital communication via CAN-BUS with other system components such as chargers, inverters, alternators, etc.



3 Interfaces

3.1 Description



3.2 Power connexion

Use Amphenol female Surlok contacts to connect the power harness to the plus and minus poles of the PowerModule, for example:

For the negative pole of the SLPPB35BSB connectors (black)

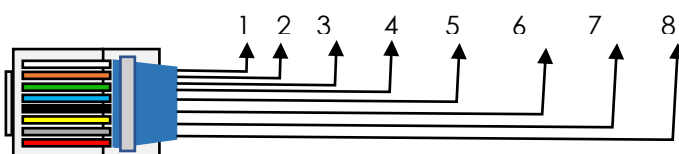
For the positive pole of the SLPPB35BSR connectors (red)

3.3 Interfaces RJ45

CAN link

The PowerModules can be controlled by CAN-Bus, the specification of the commands is described in annex 1

RJ45 connector pin-out, connector locking pin is on top.





RJ45 OUT - PINOUT

PIN#	NAME	DESCRIPTION
1	CAN H	Isolated CAN
2	CAN L	
3	Resistance 120 Ω	120 Ohms termination resistor option Connect this pin 3 to PIN 1 to have a 120 Ohms termination
4	GND	Isolated GND CAN
5	LIFELINE1	Optional use in assemblies
6	ON/OFF iso +	ON/OFF isolated Apply a voltage of +5V to 12V between Pin7 and Pin 6 to turn on the battery (external switch option to connect 6 and 7)
7	ON/OFF iso -	
8	Wake UP_IN	Used to start the next battery in assemblies
shield		Isolated GND CAN

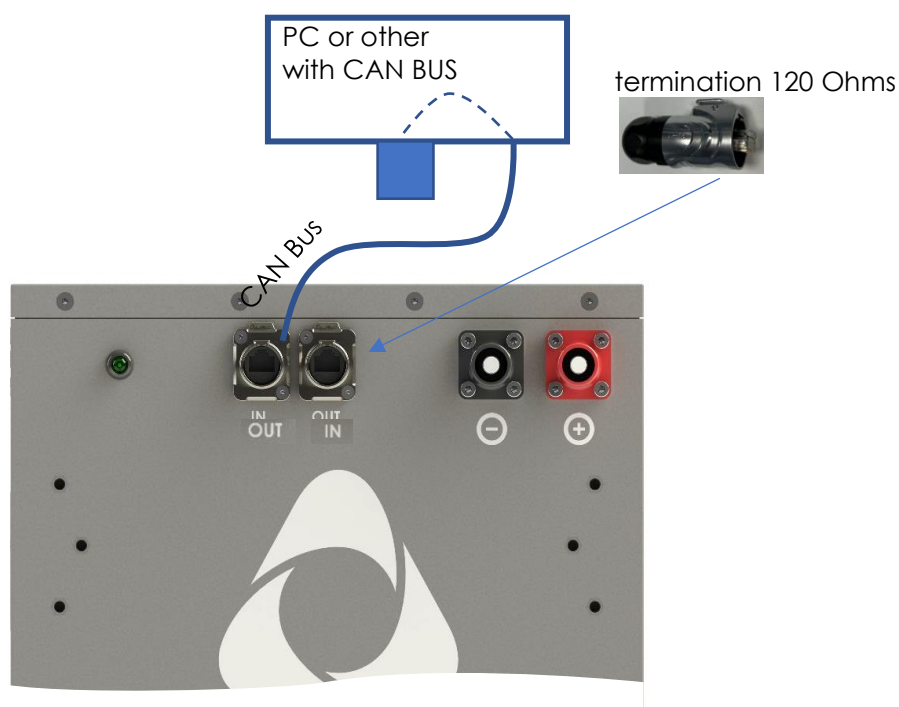
A single module can be powered up with only ON/OFF supplied in 5V or 12V.

When the PowerModule is used on its own, it must be a Master, which means it is fitted with a precharge. The principle of a precharge is described in Chapter "role of the supermaster". The precharge resistor is 25 Ohms.

A Master PowerModule cannot be used in a serial configuration.

An optional power switch can be delivered upon request. The PowerModule is then modified, pin 6 and 7 of RJ45 OUT must be connected to the external switch, it can no longer be switched on by applying a voltage. This option should only be used for single or parallel batteries, i.e. less than 60V.

The CAN-BUS link can be used to exchange information with the battery; in this case, depending on the physical configuration of the CAN, the 120 Ohm termination resistor can be used.
Recommended scheme: A 120 Ohm plug at each end of the CAN bus.



RJ45 IN - PINOUT

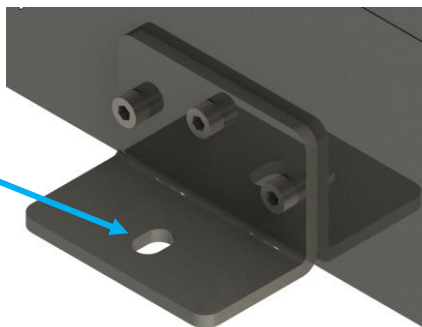
PIN#	NAME	DESCRIPTION
1	CAN H	Isolated CAN connected to RJ45 IN
2	CAN L	
3	Resistance 120 Ω	120 Ohms termination resistor option Connect this pin 3 to PIN 1 to have a 120 Ohms termination
4	GND	Isolated GND CAN
5	LIFELINE2	Optional use in assemblies
6	NC	
7	NC	
8	Wake UP_OUT	Used to start the next battery in assemblies
shield		Isolated GND CAN

3.4 Connection of chassis to ground

The ground wiring is installed with a ground braid (not supplied) connected to the system mounting bracket. The type and characteristics of the braid must meet the standards in force in the country of installation. The braid must connect the chassis to ground in accordance with the country's regulations.



Connect to ground (M8 screw)



3.5 LED Lightning code

Mode	Condition	LED
Stand-by	-	OFF
Warning	-	Steady Orange
Error	-	Steady Red
Nominal	$0\% \leq \text{SOC} < 15\%$	Orange flash 0.5Hz
	$15\% \leq \text{SOC} < 100\%$	Green flash 1Hz
	Manually opened contactor	red flash 0.5Hz
	Chauffage actif	Steady Green

An error stops the battery: its internal power contactor opens.

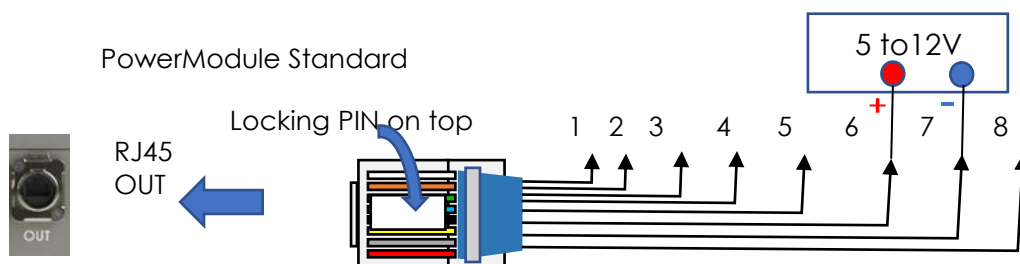
A warning does not stop the battery, it indicates a minor anomaly.

3.6 Commissioning of a single PowerModule

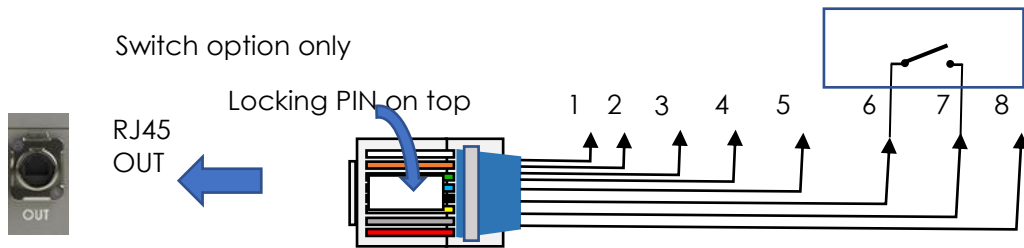
The PowerModule does not present any voltage at the power terminals without a power-up.

A PowerModule alone needs only one signal for power up, this is the ON/OFF signal described in 3.3: in the RJ45 OUT connector two contacts are used:

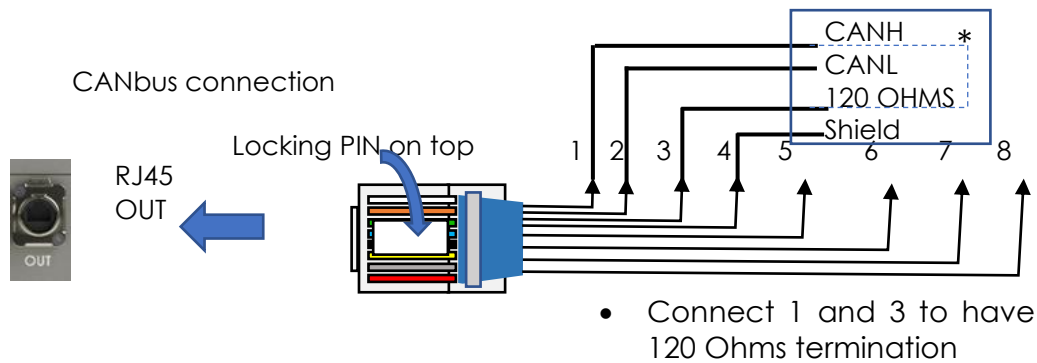
Apply a voltage of +5V to 12V between Pin 7 (minus) and Pin 6 (plus) to power up the battery. These contacts are insulated and therefore completely independent of the battery voltage.



There are PowerModules with an external switch option, simply connect Pin 6 and Pin 7, this option should not be used in serial assembly as the potential of the contacts is connected to the battery potential.



A CANbus link is provided, this link gives all the internal information of the battery. Users will benefit from using this information in their system, in particular to get the state of charge of the battery (SOC). The wiring is described in 3.3 "RJ45 OUT connector". The CAN link requires terminating resistors, in practice a 120 Ohm resistor at each end of the line; the PowerModule has an internal resistor that can be activated. CANbus connection is insulated.



The CAN commands are described in this document in the chapter "CAN bus Power Module protocol".



4 Assembly of PowerModules without SuperMaster

Doing without a SuperMaster is only valid in parallel assemblies.

The number of PowerModules in parallel must be initialized by a software parameter setting in the PowerModule Master so that it can synthesize the current.

4.1 Parallel assembly

The modules can be combined in parallel with a maximum of 31 modules in parallel (31P).

A Master PowerModule must be used; it is switched on by an external voltage or by an optional pushbutton, and switches on the other PowerModules.

IMPORTANT: the "Plus" power cables between PowerModules and power hub must all be the same length, as must the "Minus" cables.

With the same aim of balancing the current in each parallel branch, the connections in the power hubs must be carefully made.

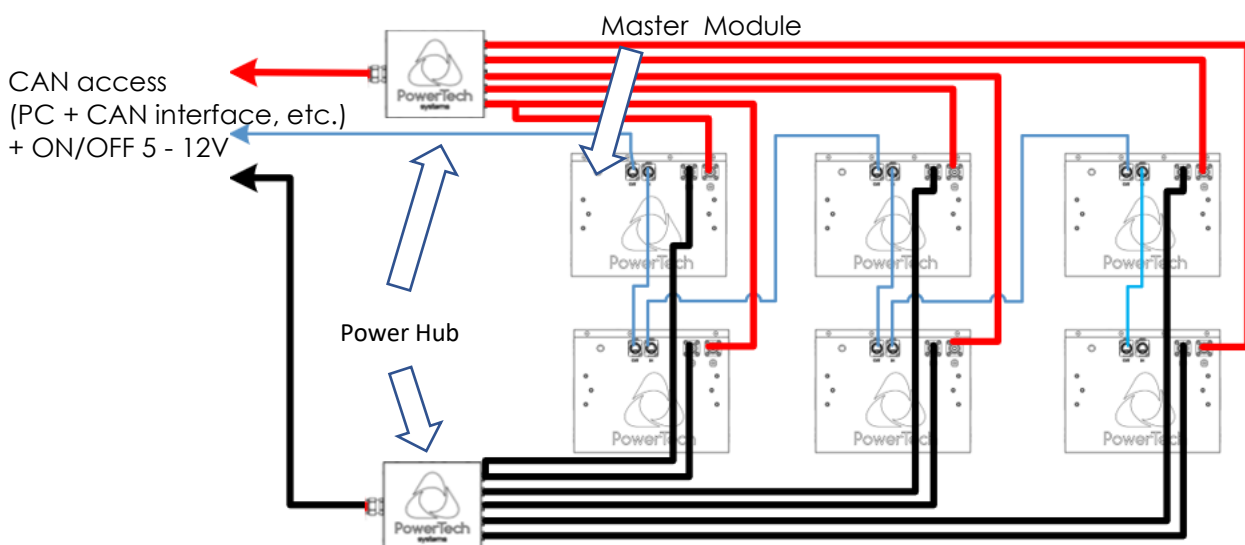
Example of 6 modules in parallel without SuperMaster:

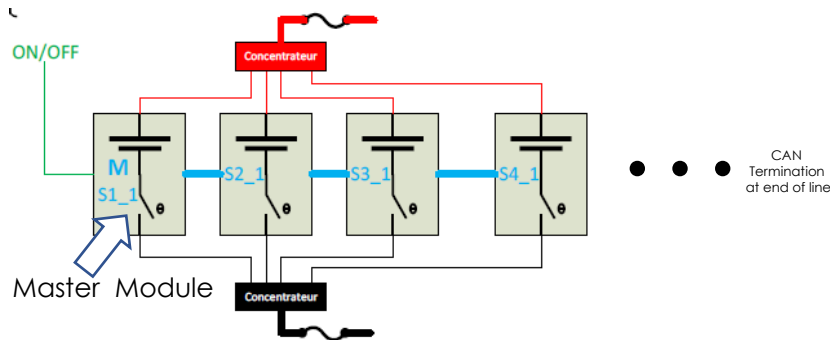
This assembly constitutes a battery with the following characteristics:

Voltage : 6 modules in parallel, i.e. 51.2V nominal.

Capacity : $6 \times 105\text{Ah} = 630\text{Ah}$

Energy : $6 \times 5,376\text{kWh} = 32,256\text{kWh}$





For parallel assembly use power link hubs.

Fuses must be installed on the negative and positive battery poles.

They must be suitable for use to protect devices, cables or connections, and lower than the internal fuse rating of the PowerModules which is 250A, so if P batteries are in parallel, the fuse must be lower than P times 250A.

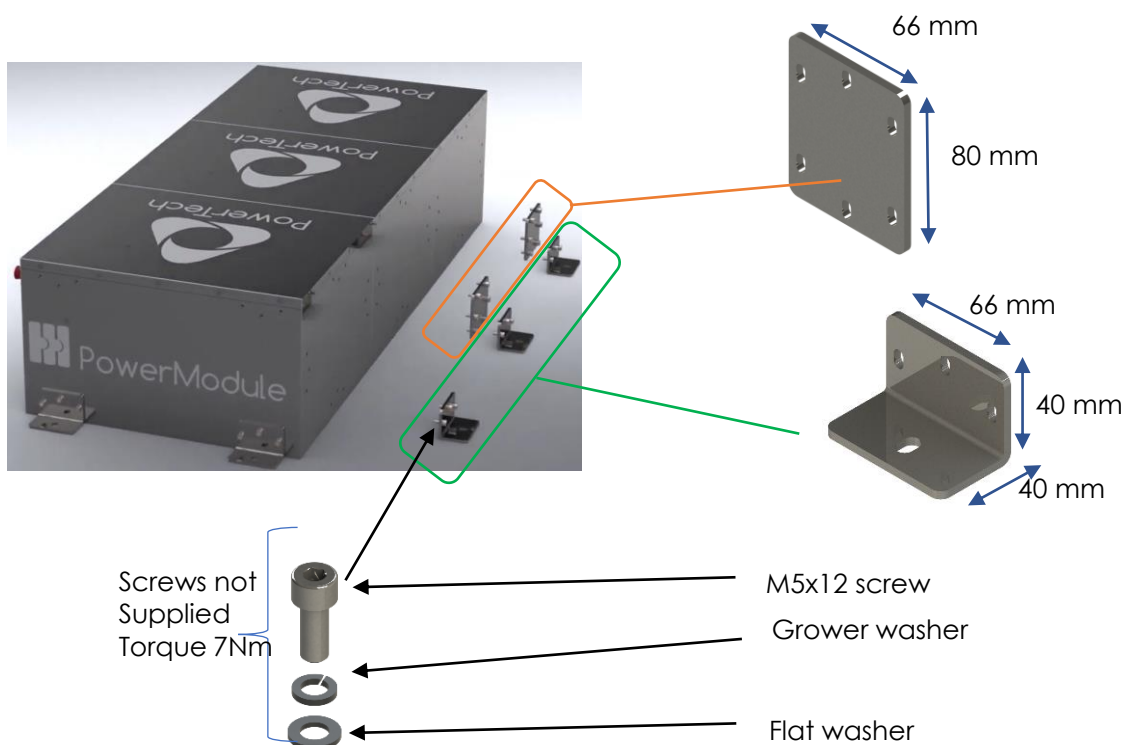
Switch start option:

ON/OFF by switch is possible. The PowerModules Master can be specifically manufactured with this option upon request. In this case the PowerModules cannot be switched on by an external voltage because the switch is not isolated from the battery voltage. This option should be reserved for 1SxP assemblies, that means only parallel assembly (51.2V nominal).

5 Mechanical assembly of the PowerModules

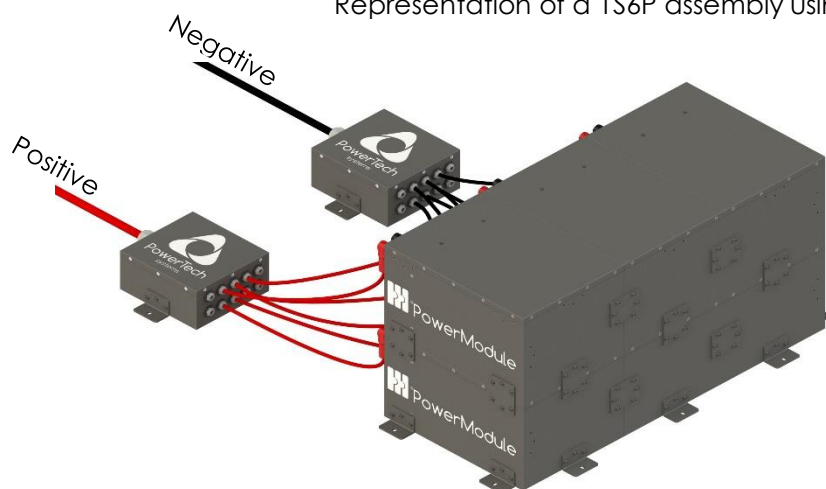
The PowerModule is easily installed and fits in any space, so there is no need for a special cabinet. The modules can be assembled with screwed plates.

PowerTech Systems supplies the assembly parts for the PowerModules.

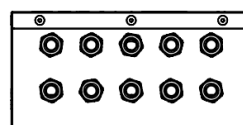
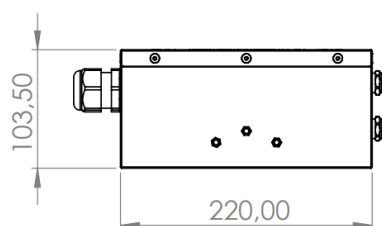
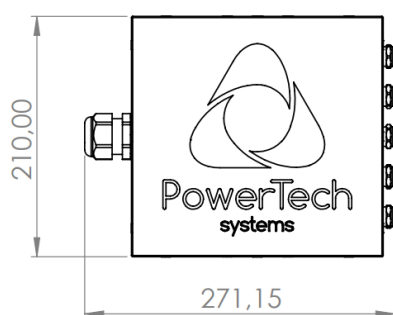




Representation of a 1S6P assembly using PowerHub

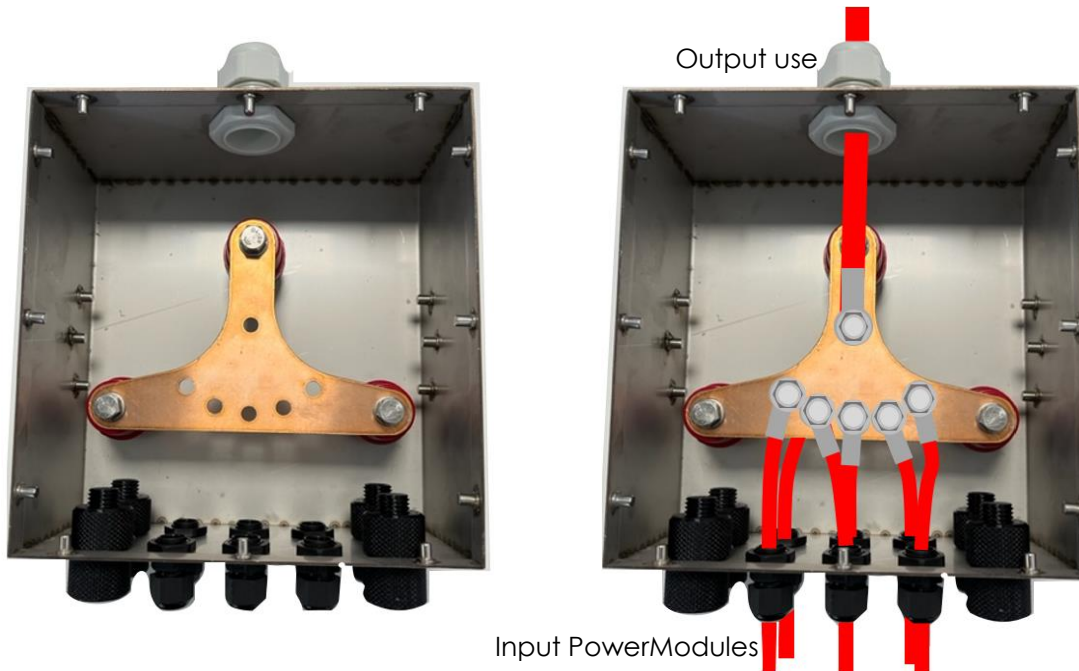


PowerHub Dimensions:



Maximum diameter of the ten input cables 9mm, 20mm output.

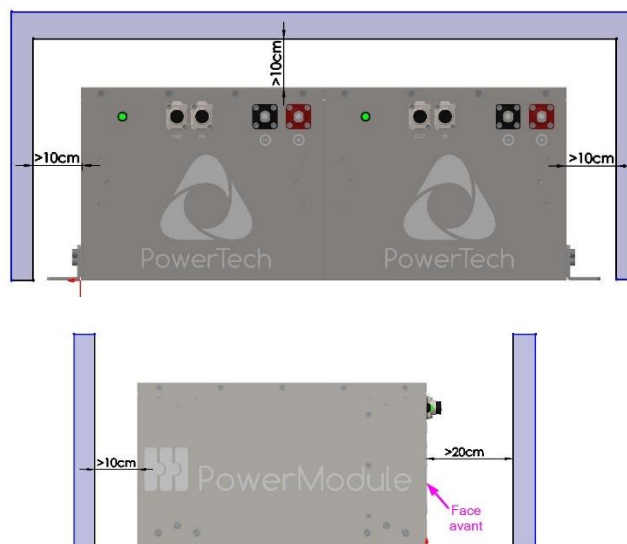
To wire a PowerHub, it is necessary to open it with a CR-V T20 torx key



To secure cables with suitable cable lugs, use M8 bolts with washers and lock washers. Tighten to 8.2Nm. Close the cover by tightening the 12 screws to 2.4Nm.

5.1 Installing several PowerModules

In systems with several PowerModules and where the current does not exceed C/2 outside peak currents, the following wall distance is recommended:



In the case of a closed housing, top and bottom ventilation must be provided.
For higher currents, PowerTech Systems should be consulted.

note: PowerModules can be handled and positioned using a suction cup:



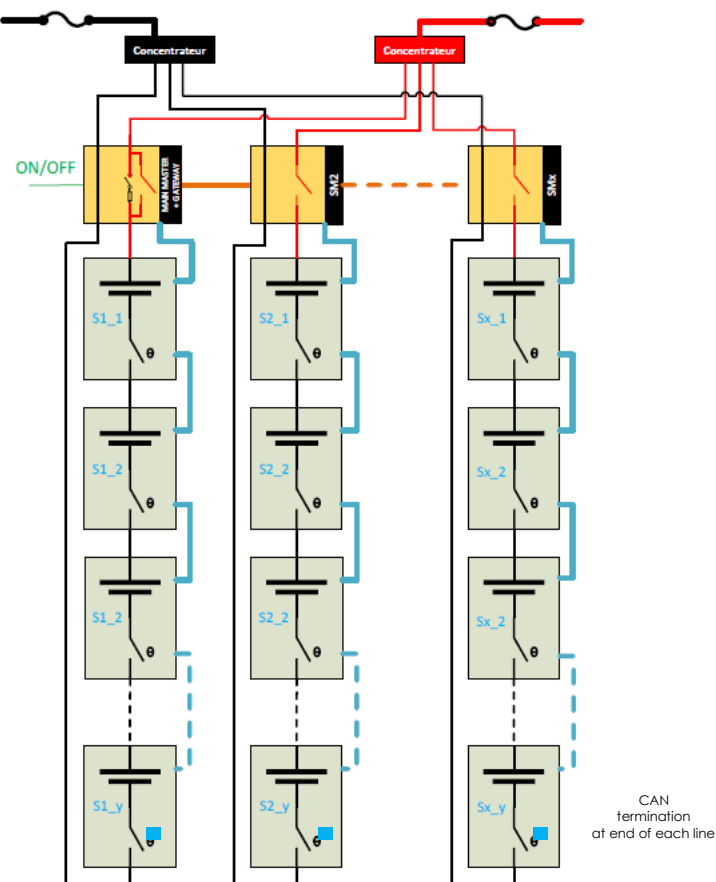


6 Assembling PowerModules with SuperMaster

The use of a SuperMaster module with the PowerModules allows to reach high voltages and powers.

The MainMaster is the SuperMaster that allows communication with the outside world and is equipped with a preload.

General wiring principle Y in series, X in parallel:



Example of calculation of assembly of 4 modules in series and 3 in parallel with their 3 SuperMasters:

This assembly constitutes a battery with the following characteristics:

Voltage: 4 modules in series, i.e. $4 \times 51.2V = 204.8V$ nominal.

Capacity $3 \times 105Ah = 315Ah$

The energy is $4 \times 3 \times 5,376kWh = 64,512kWh$

Larger assemblies require one or more SuperMasters.

The SuperMaster is an independent management module that does not include battery cells, it manages a serial battery line through "vertical" CAN Bus links.

A "horizontal" CAN bus link connects the SuperMasters together.

Fuses must be installed on the negative and positive battery poles.



They must be suitable for use to protect devices, cables or connections, and lower than the internal fuse rating of the PowerModules which is 250A, so if P batteries are in parallel, the fuse must be lower than P times 250A.

PowerModules assembly capabilities with SuperMaster :

Up to 128 modules can be assembled to reach a nominal capacity of 688kWh.

Possible assemblies:

1S (51.2V): from 1 to 128P
2S (102.4V) : from 1 to 64P
3S (153.6V) : from 1 to 42P
4S (204.8V) : from 1 to 32P
5S (256.0V) : from 1 to 25P
6S (307.2V) : from 1 to 21P
7S (358.4V) : from 1 to 18P
8S (409.6V) : from 1 to 16P
9S (460.8V) : from 1 to 14P
10S (512.0V) : from 1 to 12P
11S (563.2V) : from 1 to 11P
12S (614.4V) : from 1 to 10P
13S (665.6V) : from 1 to 9P
14S (716.8V) : from 1 to 9P
15S (768.0V) : from 1 to 8P
16S (819.2V) : from 1 to 8P

6.1 Role of the SuperMaster

Role	Description
ON/OFF Global	ON/OFF function by contact between 2 contacts of the XLR connector located at the back of the SuperMaster.
BATTERY STACK or SYSTEM MANAGEMENT	Communication with SuperMaster dependent slaves, Calculation of the global SOC (state of charge) and SOH (state of health) of the associated slaves Communication with adjacent SuperMasters.
PRECHARGE (Optional)	A SuperMaster can handle the optional preload.
INTERNAL SYSTEM COMMUNICATION	The SuperMasters communicate with each other to provide global battery information over CAN bus.
EXTERNAL SYSTEM COMMUNICATION	As an option, a gateway allows to transmit information on the WEB for a real time monitoring.



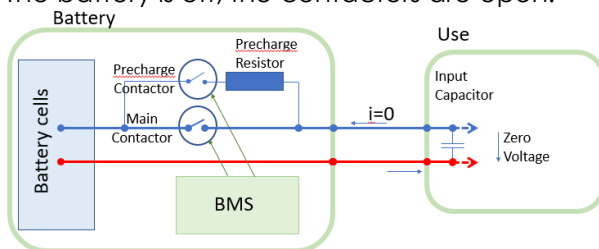
Precharge :

Precharge is used to limit the current draw on capacitors. When the battery is powered up, before the final power up, the positive pole is connected to the battery through a resistor inside the SuperMaster. The capacitors are thus charged with a limited current.

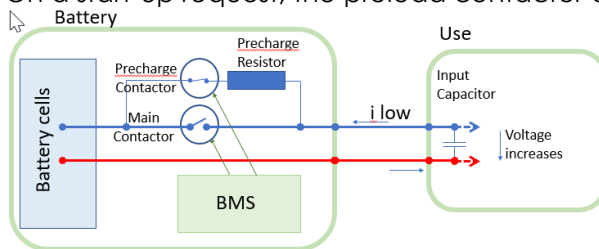
The main contactor of the MainMaster is thus preserved, when it is closed, the voltage at its terminals is limited.

Operating details:

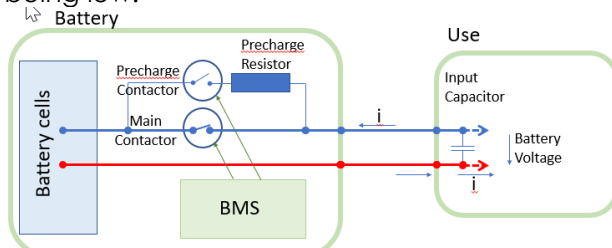
The battery is off, the contactors are open:



On a start-up request, the preload contactor closes. The current i is limited by the preload resistor:



When the voltage is sufficiently high, the main contactor can close, the voltage at its terminals being low:



The value of the precharge resistance depends on the battery voltage, the value of the capacitors and the precharge time. The user must know the value of these capacitors to adapt the resistance to the system.

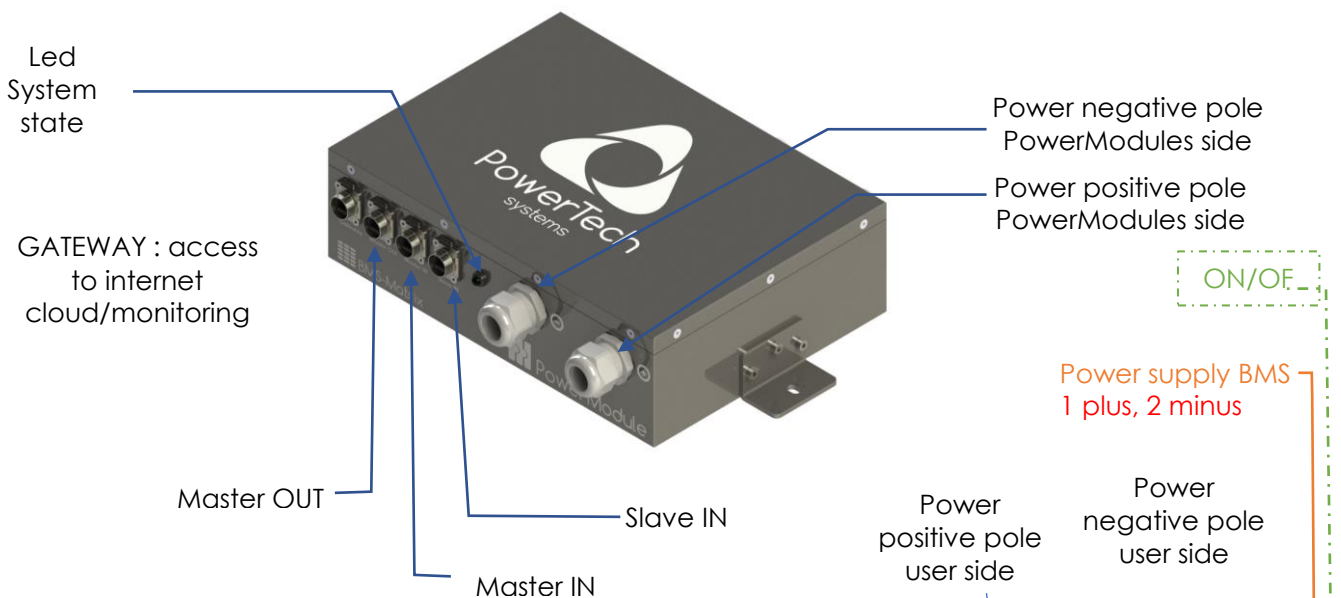


6.2 PowerModule "SuperMaster"

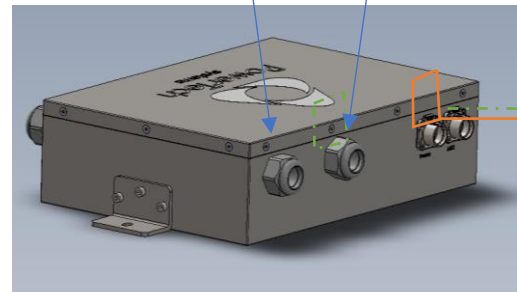
The PowerModule SuperMaster unit is smaller than the main PowerModule unit. This unit has no cells on board, only electronic control, digital communication and power management. The SuperMaster can play 2 distinct roles depending on the application and the amount of energy required. These roles are:

- **SuperMaster**
- **MainMaster** : it is a particular SuperMaster of the system, it concatenates the data of the SuperMasters of the system to deliver the global data of the battery.
In multi-Stack systems (several branches in parallel), **MainMaster** is SuperMaster n°1

6.2.1 Physical description of SuperMaster



Front view of SuperMaster

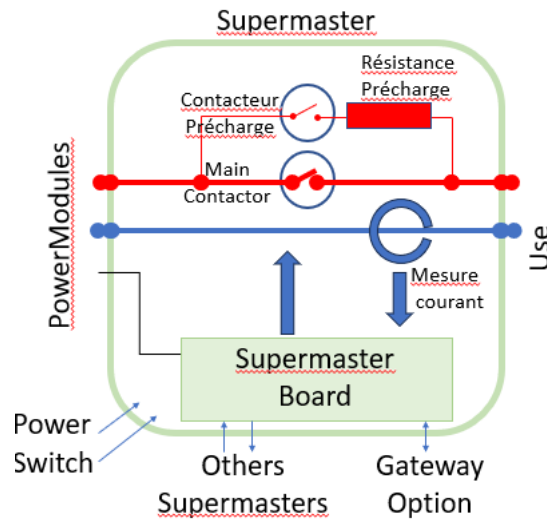


Back view of SuperMaster

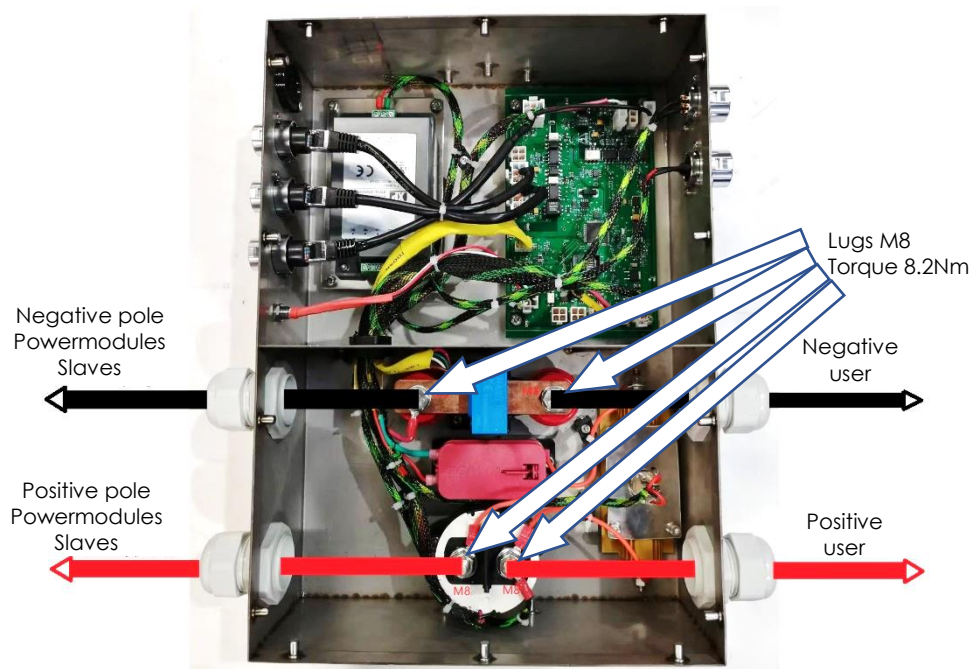
- Slave IN : CAN port RJ45 to connect the CAN link coming from a slave module.
- Master IN : CAN port RJ45 to connect the CAN link coming from a master module.
- Master OUT : RJ45 CAN port to connect the CAN link to another master module.
- Gateway: RJ45 port providing a communication interface with protocols other than CAN.
- DEL : Light-emitting diode indicating the operating status of the system or module.



- XLR 2-pin "Power" connector is used to power SuperMaster with 12V, or 24V or 51.2V according to the option: Pin 1 plus, Pin 2 GND.
- XLR 7-pin connector "AUX" is used to wake up the SuperMaster with a dry contact between Pin 1 and Pin 2. The other pins will be used for future applications.



Internal SuperMaster Wiring



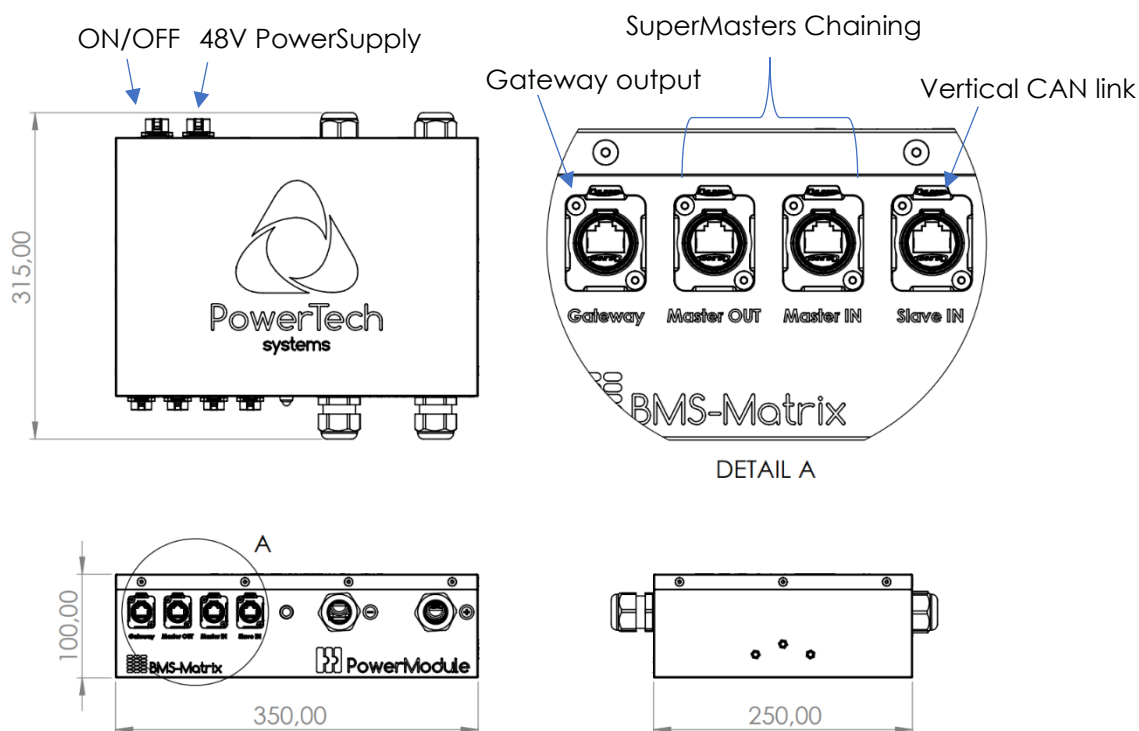


6.2.2 LED lightning code

Mode	Condition	LED
Stand-by	-	OFF
Warning	-	Steady Orange
Error	-	Steady Red
Nominal	$0\% \leq \text{SOC} < 15\%$	Orange flash 0.5Hz
	$15\% \leq \text{SOC} < 100\%$	Green flash 1Hz
	Manually opened contactor	red flash 0.5Hz
	Chauffage actif	Steady Green

An error stops the battery : its internal power contactor opens.
A warning does not stop the battery, it indicates a minor anomaly.

SuperMaster dimensions





Supermaster characteristics

Supply voltages according to options	12.0 V / 24 V / 48 V
Consumption without gateway	340mA/160mA/128mA
Consumption with gateway	540mA/260mA/80mA
Weight (+/- 3 %)	4,9 Kg with Gateway
Dimensions (l x w x h)	350 x 250 x 100 mm
Operating Temperature	de -20°C, à +60°C
Power Connector	M8, cable diameter 20mm
continuous discharge current	400 A
Orientation	No orientation sensitive elements
Degree of protection	IP40*

*(IP54 on demand: different mechanical characteristics)



RJ45 Slave IN used in assembly

PIN#	NAME	DESCRIPTION
1	CAN H	Isolated CAN
2	CAN L	
3	NC	
4	GND	Isolated GND CAN
5	LIFELINE	Optional use in assemblies
6	NC	
7	NC	
8	Wake UP Slave	Used to start the first PowerModule in assemblies

A straight RJ45 cable is used in this socket

RJ45 Master IN utilisé used in assembly

PIN#	NAME	DESCRIPTION
1	CAN H	Isolated CAN
2	CAN L	
3	Resistance 120 Ω	120 Ohms termination resistor option Connect this pin 3 to PIN 1 to have a 120 Ohms termination
4	GND	Isolated GND CAN
5	LIFELINE	Optional use in assemblies
6	ON/OFF iso	Can be use but ON/OFF preferred
7	GND	
8	Wake UP_IN	Used to start the SuperMaster in assemblies

A straight RJ45 cable is used in this socket



RJ45 Master OUT used in assembly

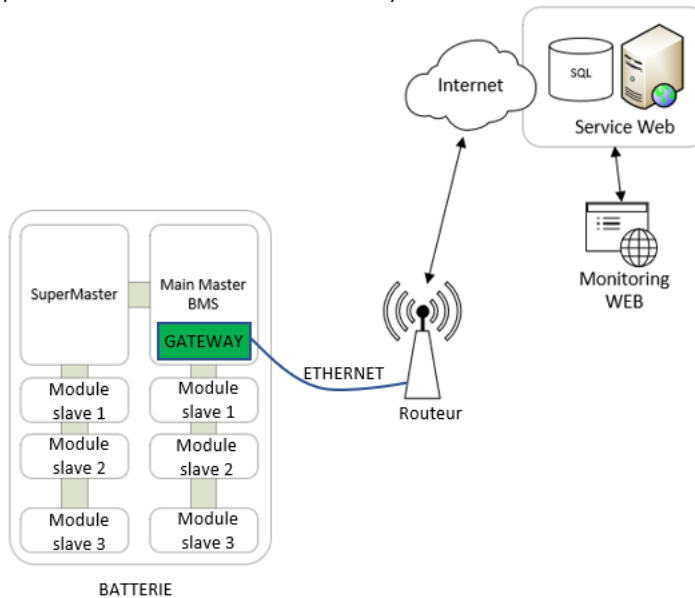
PIN#	NAME	DESCRIPTION
1	CAN H	Isolated CAN
2	CAN L	
3	Resistance 120 Ω	120 Ohms termination resistor option Connect this pin 3 to PIN 1 to have a 120 Ohms termination. There is also an option jumper on the SuperMaster board
4	GND	Isolated GND CAN
5	LIFELINE	Optional use in assemblies
6	NC	
7	NC	
8	Wake UP_OUT	Used to start the Next SuperMaster in assemblies

A straight RJ45 cable is used in this socket

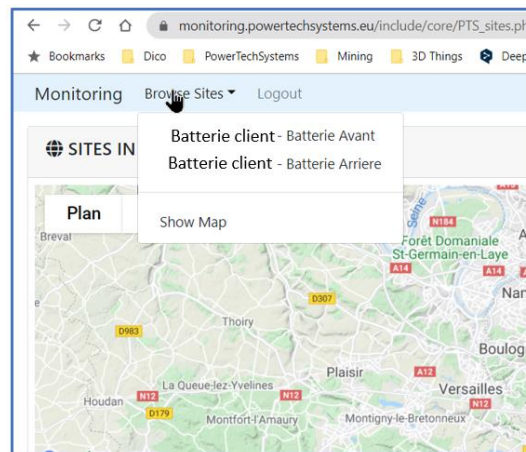


6.2.3 Gateway and monitoring option

In a PowerModule system, the MainMaster can be equipped with an optional Gateway card. In combination with a router, it provides remote access to the battery parameters. A monitoring system allows simple presentation of these parameters. The router uses the SuperMaster's Ethernet "Gateway" link.



Each battery system in the example "Avant" and "Arriere" has its own database. The user selects the system in the "Browse Sites" menu.

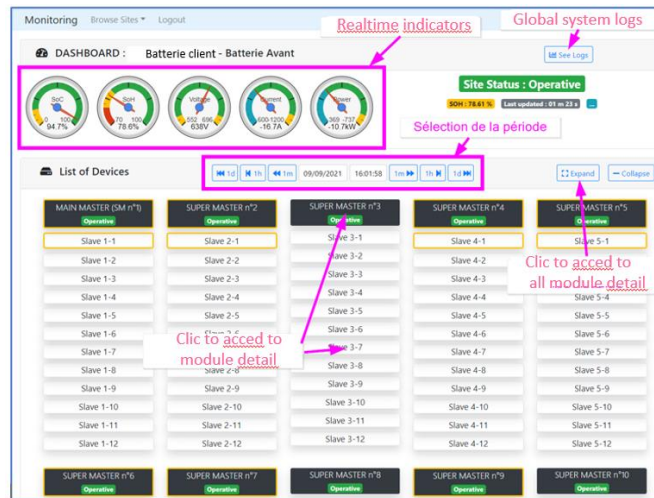


The main monitoring screen shown below displays all the system data. It is possible for the user to select a specific date and time to see the status of the system at that moment.

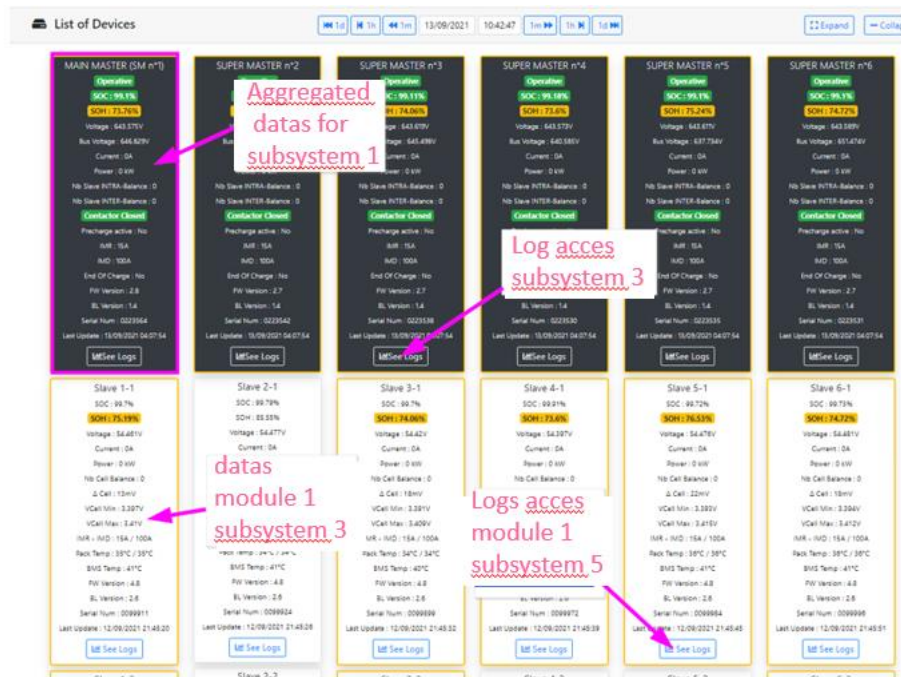


PowerModule

Installation and user manual



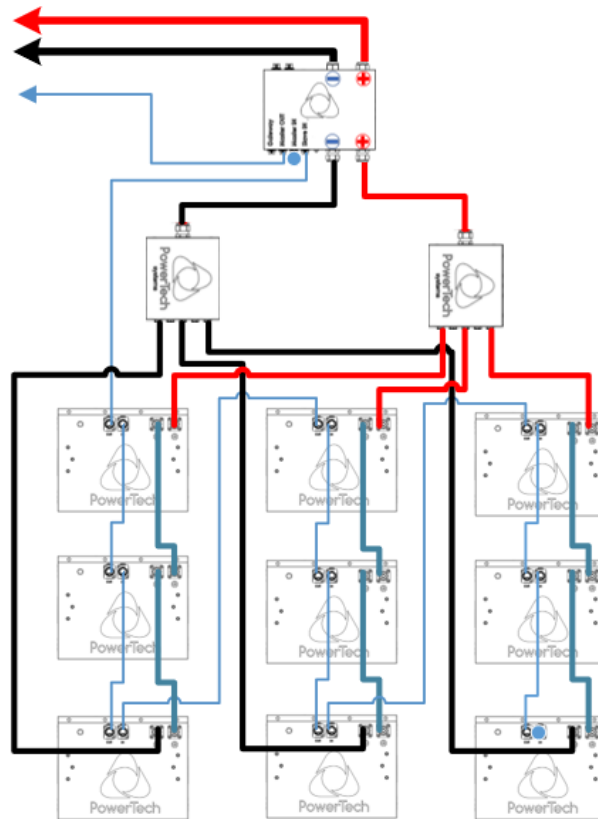
The information is available for all SuperMaster BMS (black background) and all battery modules (white background).



Des logs graphiques peuvent être présentés sous forme de graphe :



29





7 Commissioning of PowerModule system

To start up Make the connection as shown in the different examples,

If there is no Supermaster, the MainPowerModule will be switched on as described in the chapter "Commissioning a single PowerModule". The chaining of the other PowerModules allows them to power up. The switch-on option can only be used when the PowerModules are in parallel.

If there is at least one SuperMaster, the MainSuperMaster switch will power up the whole system through the RJ45 chaining of the PowerModules and SuperMasters.

Wire the PowerModules and the inside of the SuperMasters as described above.

Connecting the power to all the SuperMasters

If there are several Supermasters, connect the switch to the MainSuperMaster only.



Switch
Connector 2 points recommended :
LP-20-C02PE-01-22
of CNLINKO

Power
12V or 24V or 51,2V according option
Connector 7 points recommended :
LP-20-C07PE-01-22
of CNLINKO
1 plus, 2 minus

7.1 First commissioning and charge

Important for **initial start-up**_:

An initial charge must be carried out before commissioning. At the end of this charge, all the PowerModules must have reached the End of Charge state. This step is important for the battery to work properly.

Full charge occurs when the END OF CHARGE message is presented on the CANBUS network: Status 2= End of Charge System. Each charge should end with this signal. End of Charge is also indicated by a steady green LED from software version 5.11 for PowerModules and version 3.5 for SuperMaster..

The End of Charge indication is shown when the system is charged and balanced.



Charging and balancing:

Under certain conditions, PowerModules can become unbalanced.

This imbalance corresponds to a voltage difference between cells in series inside the PowerModule. It can occur during prolonged storage or if charges are not complete.

Consequences of an unbalanced battery:

- The full capacity of the battery cannot be used.
- The battery cuts out prematurely when charging, i.e. at too low a voltage.
- The battery cuts out prematurely when discharging, i.e. at too high a voltage.

How to balance

Balancing takes place at the end of the charge. It is only fully effective from a charging voltage of 56.0V.

The recommended charging voltage for a PowerModule is 57.0V. The END OF CHARGE message is displayed when the battery is charged and balancing is satisfactory. Consequently, each charge must be complete. By respecting this instruction, the battery will maintain its performance.

Balancing takes place at the end of the charge. It is only fully effective from a charging voltage of 56.0V.

The recommended charging voltage for a PowerModule is 57.0V. The END OF CHARGE message is displayed when the battery is charged, and balancing is satisfactory.

If the charger is unable to modulate its charging current from the IMR, charging will probably result in relay opening/closing until the end of charging and balancing. This mode of operation is normal and is designed to protect the battery from overloads linked to the charger, which does not manage the IMR setpoint. Charging and balancing without IMR takes longer.

A very large imbalance could require several days' charging to restore nominal battery operation.

Another method of balancing a PowerModule is to make a normal load up to 55V and then finish the load with a low current of 0.4A up to 57V, which can be achieved with a stabilized power supply.

Consequently, each charge must be complete. If you follow these instructions, the battery will maintain its performance over time.

Floating charge

A balanced charged PowerModule can be kept operational over a long period using a charger with a floating voltage of 53.4V. The PowerModule will remain perfectly charged for a long period, self-discharge will be compensated and aging will be reduced.



8 Maintenance

The PowerModule System does not require any special maintenance. It is necessary to check the electrical installation at least once a year. Defects such as loose connections, damaged or burned cables, etc. must be corrected immediately.

Storage / wintering

All modules should be switched off before being stored in a dry, well-ventilated place. The self-discharge rate is less than 5% per month. This rate and the natural ageing of the battery increases with the ambient temperature.

If a Lithium-Ion battery is left unused for more than one month, we recommend :

- Charge the battery before storage. The charge level for storage should be 100%
- Check the state of charge every three months and recharge occasionally if necessary.
- **The battery should never be left fully discharged for more than 7 days** as this will degrade the cell chemistry and shorten the life of the battery.
- Calendar ageing: Batteries age over time even without use. Avoid prolonged exposure to heat, a regulated temperature of 15-25°C is ideal. Higher temperatures accelerate ageing.

The storage room must be dry and ventilated

IMPORTANT

The on-board electronics (BMS) has a residual but continuous energy consumption to ensure its operation. If the battery is not used for a long period of time, it is necessary to switch off the battery to prevent the battery from being slowly discharged.

If the battery is left on for too long without recharging, the BMS will automatically switch off at a set voltage threshold to prevent damage to the cells by deep discharge. However, the on-board BMS continues to discharge the battery cells very slowly.

NEVER STORE A DISCHARGED OR SWITCHED ON POWERMODULE SYSTEM.

Replacing a module:

The replacement of a module cannot be done without precautions:

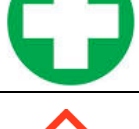
The first one concerns general safety precautions: see the chapter on general precautions.

The second concerns the state of charge: When using the new PowerModule, it must reach 100% at the same time as the others. The best solution is to charge the whole battery to 100% before the change and then to make the change with the new PowerModule independently charged to 100%. The battery will then be balanced and can deliver the maximum capacity..



9 General precautions

9.1 Main rules

	Follow these instructions and keep this manual within reach, not far from the Lithium-Ion battery for future reference. Work on the Lithium-Ion battery should only be carried out by qualified personnel.
	Danger of explosion or fire. Avoid short-circuiting power connectors. Prohibit deep discharges or high charge and discharge currents. Refer to the technical limitations of the battery later in this document. In case of fire, use a D-foam or CO2 extinguisher.
	ELECTRICAL HAZARD: The PowerModules have a nominal voltage of 51.2V and a maximum voltage of 58.4V. These products are classified as Extra Low Voltage and should only be handled and installed by trained and qualified personnel. Connecting PowerModules in series presents a risk of electrification, electrocution or electric shock. This operation must be carried out ONLY by trained and competent personnel. The installation and handling of a PowerModule system assembled in series must be carried out by authorized personnel with LV (Low Voltage) certification in accordance with the NF C 18-510 standard.
	When working inside the battery (to be carried out only by authorised personnel), it is necessary to wear safety glasses, gloves and protective clothing.
	Splashes of battery materials such as electrolyte or powder on the skin or eyes should be washed off thoroughly with clean water. It is necessary to consult a doctor afterwards. Rinse off any splashes on clothing with clean water. Under normal conditions of use, contact with this material is impossible.
	Never attempt to open or disassemble the PowerModule product. Electrolyte is a highly corrosive material. Under normal operating conditions, contact with the electrolyte is impossible. If the battery casing is damaged and electrolyte leaks, do not touch exposed products which are aggressive to the skin and eyes.
	Lithium-Ion PowerModule batteries can be charged with a voltage of up to 58.4V. They can be discharged to 44V. Note that this voltage range is wider than you would find for other types of batteries, such as lead-acid batteries. These voltages may exceed those supported by the connected equipment. Therefore, measures must be taken to protect the powered equipment.
	The warranty shall be void if the general rules are not observed, if the instructions for use are not followed, if work is carried out on the product or if the products are opened without prior permission.



9.2 Transport precautions

	The PowerModule lithium-ion battery must be transported de-energised (powered down and all power connectors disconnected). The power connectors must be electrically isolated (plastic cover on each power connector) The PowerModule lithium-ion battery must be transported in its original packaging. Do not use the plugs to lift the battery. The batteries belong to UN3480 and UN3481, Class 9, Packaging Group II and must be transported and shipped in accordance with these regulations. This means that for land or sea transport (ADR, RID and IMDG) they must be packed according to P903 instructions. For air transport (IATA), instructions P965 apply.
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9.3 Recycling of Lithium-Ion batteries

Batteries marked with the recycling symbol should be taken to a recognised recycling agency. They can be returned to the distributor of the product.

With prior agreement, they can be returned to the manufacturer. Do not mix batteries with household or industrial waste.

	 Li-ion
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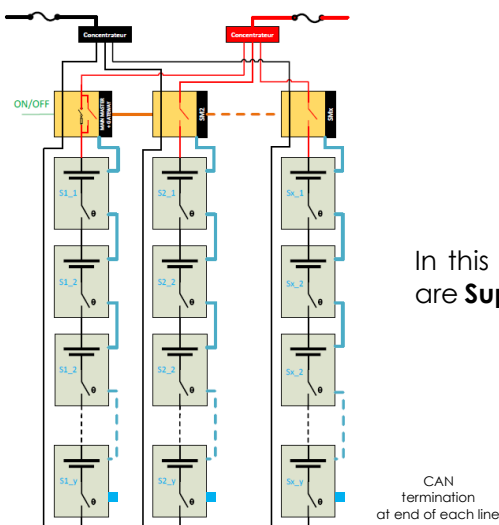
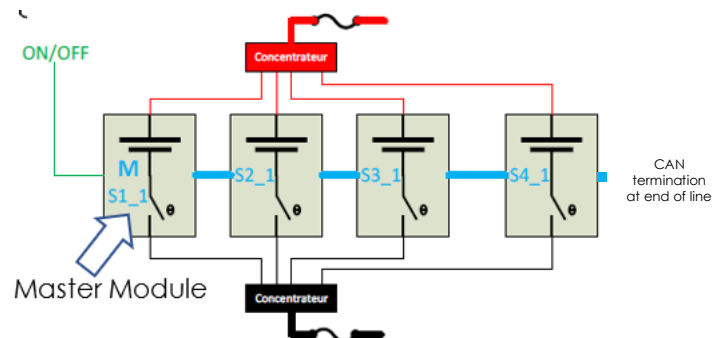


10 CAN protocol

The following describes the communication protocol used by the BMS (Battery Management System). The underlying bus protocol is CAN 2.0b.

Reminder of the terms Slave, Master, SuperMaster and MainMaster:

In this configuration, M is the **Master**
The other modules noted **S** are **Slaves**



In this configuration, the Modules noted **S** are **Slaves**, **SM** are **SuperMasters**, the SuperMaster n°1 is called **MainMaster**

10.1 Errors / Warnings

Both the master and slave batteries can generate Errors and Warnings from the battery to protect the battery. The error and warning information is listed below (refer to CAN bus message 0x022 for details):

A Warning has no effect on the battery. It is information to warn the user. But some errors may affect the power contactor (see Appendix IV Errors / Warnings and Power Contactor).



10.2 CAN bus Protocol

CAN is the acronym for "Controller Area Network".

In a CAN network, many short messages carrying brief information about the status of the nodes are broadcasted to the whole network, which ensures the consistency of data in each node of the system. Standard CAN uses an 11-bit identifier and Extended CAN uses a 29-bit identifier.

Bus CAN Specifications :

CAN Spec.:	CAN 2.0B (compatible with CAN 2.0A)
Bit rate:	125kbit/s, 250kbit/s (Default), 500kbit/s, 1Mbit/s
Message Type:	Standard (11-bit Identifier) & Extended (29-bit Identifier Default)
Remote Frame:	not use
Data Format:	Intel

10.2.1 Identifiers

The table below shows the structure of the CAN bus identifiers with the extended format. PowerTech uses the extended identifier format for its communication protocol.

sid												eid																	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
					8bits								8bits								8bits								
Prio rity	Not used		Device		Transmitter				Module number				Stack number				Frame type*		Command ID**										
0: high priority 1: low priority			10: Battery 11: Tool		01: SuperMaster or Main Master 10: Slave 11: Master 00: Broadcast				01: module 1 02: module 2 max :31 modules				01:Stack 1 02:Stack 2 max :31 stacks				00: Tx 01: Rx 10: Dreq		0x020 status 1 slave 0x021 status 2 slave ...										

CAN bus identifiers

*Note: Tx: A simple message; Rx: Request to write data; Dreq: Request to send data.

Below are some sample messages to understand the message structure of the Powertech CAN protocol:

**More details in the next chapter

The table below shows some examples of CAN messages to understand the structure of the identification.



Every object has a specific numbering from its position in the system.

The PowerModules in series makes part of the same stack which has a numbering from 1 to n (n maximum=31) (stack number), each PowerModule depending on its position in the stack has a numbering from 1 to n (n maximum=31) (module number).

The supermasters have module number 0, the global information concerning all PowerModules is transmitted with stack number 0.

The masters have module number 0, as they deliver global information as master, they have for module 1 if they deliver information as a slave.

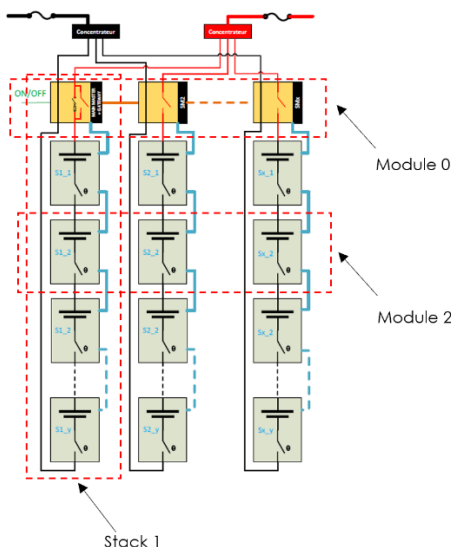
This numbering allows you to know where the information comes from or what elements are concerned by the transmitted information.

Examples of the construction of the ID:

ID messages	Description
0x02821022	Error/Warning Slave 1 in stack 1 priority 0 (High priority)

Construction of the ID

Priority	high	0b0
Device	battery	0b10
Transmitter	slave	0b10
Module number	1	0b00001
Stack number	1	0b00001
Frame type	message simple	0b00
ID message	error and warning	0x022 (on 10bits)



	Byte 3								Byte 2								Byte 1								Byte 0									
bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
					P				Trans		module				stack				frame		Command ID													
0b	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0
0x	0				2				8				2				1				0				2				2					

So, we have in hexadecimal 0x02821022



Other examples of the construction of the ID

	ID	Description
ex1	0x12C21020	Status 1 of Master with priority 1 (low priority)
ex2	0x12841020	Status 1 of Slave 2 in stack 1 with priority 1 (low priority)
ex3	0x13041A05	Maintenance message priority 1 (Low priority) Tool request cell voltage of Slave 2 in stack 1

Be careful in example 3 in the first half of Byte 1 The message ID 0x205 and the frame Dreq=0b10 combine. We therefore obtain 0xA05.

		Byte 3								Byte 2								Byte 1								Byte 0							
		31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		P				Dev				Trans				module				stack				frame				Command ID							
ex1	0b	0	0	0	1	0	0	1	0	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
	0x	1				2				C				2				1				0				2				0			
ex2	0b	0	0	0	1	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0
	0x	1				2				8				4				1				0				2				0			
ex3	0b	0	0	0	1	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	1	0	1
	0x	1				3				0				4				1				A				0				5			

10.2.2 Command ID used for operations

The command ID allows different information to be distinguished. It is also possible to manage and monitor the storage system in a more convenient way. The control ID represents the 10 least significant bits of the CAN bus identifiers.

For a battery system, the most important messages are "Global Status 1" and "Global Status 2". They are used to obtain information about the whole system, such as voltage, current, IMR, IMD, etc.

The table below shows all the identifiers used by the BMS.

ID Message (on 10 bits)	Name	Specification
0x002	Global status 1	Periodic
0x003	Global status 2	Periodic
0x011	Numbering	On start
0x015	Master command 2	Reserved
0x016	Master command 1	Periodic
0x017	Master command 3	Reserved
0x020	Status 1	Periodic
0x021	Status 2	Periodic
0x022	Error and warning	Periodic if present (0.5s)
0x103	Cell voltage 1 - 4	On Request
0x104	Cell voltage 5 - 8	On Request
0x109	Cell voltage 9 - 12	On Request
0x110	Cell voltage 13 - 16	On Request
0x105 0x016 0x107	Error_JDD	Reserved



0x200	Parameter	Request / On request
0x203	Calibration courant 0A	Reserved
0x205	Maintenance	Request
0x206	SOC	Reserved
0x207	init dialog apv	Reserved
0x208	Calibration 2 currents	Reserved
0x209	Cell balancing	Periodic if present (1s)
0x20A	Total capacity discharge	Reserved
0x20B	Calibration cell tension	Reserved
0x20C	Contactor open - close	Request
0x20E	Option	On Request
0x20F	Bit rate CAN bus	On Request
0x21F	Total slave	On Start/ Request / On Request
0x3E9	Update	Reserved
0x3EA	Update old Gateway	Reserved
0x3EB	Soft Version	Reserved
0x3EC	Update BL	Reserved

External tools can make requests to the battery system in this way.

For example:

To manage the option saved in the BMS, the command ID 0x20E can be used.

To request an option, the frame type is DREQ = 0b10. The last three half bytes of the CAN message ID are therefore 0xA0E.

For the write option, the frame type is RX = 0b01. The last three half bytes of the CAN message identifier are therefore 0x60E.

For the response, the frame type is TX = 0b00. The last three half bytes of the CAN message identifier are therefore 0x20E.

10.2.3 Periodic messages

Periodic messages are issued at regular intervals and carry the basic information of the system

Command ID (on 10 bits)	Name	Periodic
0x016	Master command 1	1 second (Can be set-up)
0x002	Global status 1	
0x003	Global status 2	
0x020	Status 1	
0x021	Status 2	
0x022	Error et warning	Periodic if present (0.5s)

Master command is an internal periodic message intended for the slaves, it allows commands to be sent to the slaves.



10.2.3.1 Global Status 1 for Master module

ID message	Period	Data Length
0x02C00002	1s	7

The informations for the global status are module 0 and stack 0.

The informations for the global states are module 0 and stack 0:																																				
Byte 3								Byte 2								Byte 1								Byte 0												
bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
					P				Dev	Trans		module				stack				frame		Command ID														
0b	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0			
0x	0				2				C				0				0				0				0				0				2			

Global Status1 Message (sent by Master module)

Description:

This message is sent by the **Master** module to provide aggregated information of all slaves present in the system.

The information includes errors, warnings and general status, IMR (maximum charge current), IMD (maximum discharge current) and SOH (state of charge) of the system.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
SOC of global system	0	16	0.01%		System SOC
IMR of system	16	14	100mA	0 → 1638.3A	Max charge current
IMD of system	30	12	1A	0 → 4096A	Max discharge current
Status of system	42	4			0 : Idle (Internal code : #100) 1 : Operative (#101) 2 : Error (#102) 4 : Part Operative
Error of system	46	1			At least one slave has error
Warn of system	47	1			At least one slave has warning
End of charge system	48	1			End of charge system
Floating system	49	1			Floating system



10.2.3.2 Global Status 2 for Master module

ID message	Period	Data Length
0x02C00003	1s	6

bit number	Byte 3								Byte 2								Byte 1								Byte 0							
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	P				Dev				Trans		module				stack				frame		Command ID											
0b	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0x	0				2				C		0				0				0		0								3			

Global Status 2 Message (sent by Master module)

Description:

This message is sent by the **Master** module and complements the information provided by Global Status 1.

The information includes current, voltage and SOH (State of Health) of the current system.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
Current of system	0	16	100mA	-3276.8 → +3276.7A	Current total
Voltage of system	16	16	25mV	0 → 1638.375V	Voltage total
SOH of system	32	16	0.01%	0% → 100.00%	State of Health total



10.2.3.3 Global Status 1 for MainMaster module

ID message	Period	Data Length
0x02400002	1s	7

	Byte 3								Byte 2								Byte 1								Byte 0							
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				P				Dev	Trans	module				stack				frame		Command ID												
0b	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
0x	0				2				4				0				0				0				0				2			

Global Status 1 Message (sent by MainMaster module)

Description:

This message is sent by the **MainMaster (SuperMaster #1)** to provide aggregated information of all slaves present in the system.

The information includes errors, warnings and general status, IMR (maximum charge current), IMD (maximum discharge current) and SOC (state of charge) of the system.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
SOC of system	0	16	0.01%	0→100.00%	System SOC value
IMR of system	16	14	100mA	0→1638.3A	Current max charge
IMD of system	30	12	1A	0→4095A	Current max discharge
Status of system	42	4			0: Idle 1: Operative 2: Error 3: Precharge 4: Part operative
Error of system	46	1			At least one rack has error
Warn of system	47	1			At least one rack has warning
End of charge system	48	1			End of charge system



Data Decoding Example

For data decoding we use the little endian method, i.e. the least significant bit is the last, the bit numbering is as indicated in the table below.

Be careful here, the data concerning several different data can be found on the same byte, you must therefore be careful with the division.

Here we receive the message 0x0E13ACCDF38500

bit number	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8	23	22	21	20	19	18	17	16	31	30	29	28	27	26	25	24				
0b	0	0	0	0	1	1	1	0	0	0	0	1	0	0	1	1	1	0	1	0	1	1	0	0	1	1	0	0	1	1	0	1				
0x	0				E				1				3				A				C				C				D							
bit number	39	38	37	36	35	34	33	32	47	46	45	44	43	42	41	40	55	54	53	52	51	50	49	48												
0b	1	1	1	1	0	0	1	1	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0												
0x	F				3				8				5				0				0															

SOC of system		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
49,78%	0b	0	0	0	1	0	0	1	1	0	0	0	0	1	1	1	0
	0x	1				3				0				E			
	0d	4978															

IMR of system		29 28 27 26 25 24						23 22 21 20				19 18 17 16			
350 A	0b	0 0		1 1		0 1		1 0		1 0		1 1		0 0	
	0x	0		D				A				C			
	0d	3500													

IMD of system													
		41	40	39	38	37	36	35	34	33	32	31	30
1999 A	0b	0	1	1	1	1	1	0	0	1	1	1	1
	0x	7				C				F			
	0d	1999											

Status of system		45	44	43	42
Operative	0b	0	0	0	1
	0x	1			
	0d	1			

Error of system	46
	0

Warn of system	47
at least one rack has warning	1

End of charge system	48
	0



10.2.3.4 Global Status 2 for MainMaster module

ID message	Period	Data Length
0x02400003	1s	6

	Byte 3								Byte 2								Byte 1								Byte 0							
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					P				Trans	module				stack				frame		Command ID												
0b	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0x	0				2				4				0				0				0				0				3			

Global Status 2 Message (sent by MainMaster module)

Description:

This message is sent by the **MainMaster (SuperMaster #1)** and it completes the information provided by Global Status 1.

The information includes current, voltage and SOH (State of Health) of the current system.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
Current of system	0	16	100mA	-3276.8 → +3276.7A	Current total
Voltage of system	16	16	25mV	0 → 1638.375V	Voltage total
SOH of system	32	16	0.01%	0% → 100.00%	State of Health total



10.2.3.5 Error and Warning Master and Slave

ID message	Period	Data Length
0x02YYY022	500ms when error or warning	8

Note:

- "y1 y2 y3 " represents the transmitter ID, module ID and stack ID concerned by the message.

	Byte 3								Byte 2								Byte 1								Byte 0								
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
	P				Dev				Trans		module				stack				frame		Command ID												
0b	0	0	0	0	0	0	0	1	0	y1	y1	y2	y2	y2	y2	y2	y3	y3	y3	y3	y3	0	0	0	0	0	0	1	0	0	0	1	0
0x	0				2				Y				Y				Y				0		2				2						

message Error/Warning (sent by Master and Slave modules)

Description:

These messages are sent by the **Master** and **Slave** modules. They are collected to check the integrity of the stack.

This message is sent when there is at least one error or warning. As soon as they are corrected, the BMS sends this message twice with the data 0.

Data information:

Data name	Internal Event Code	First Bit	Length	Comments
Error Output Current	#1	0	1	Rise when output current > 300A or when output current > 260A during 30s, or when output current > 200A during 300s, Value max output current can be set-up
Error Input Current	#2	1	1	Rise when input current > 110A or when input current > 105A during 30s, Value max input current can be set-up
Error Cell Over voltage	#3	2	1	Rise when one cell > 3850mV, This value can be set-up
Error Cell Under voltage	#4	3	1	Rise when one cell is < 2700mV, This value can be set-up
Error Cell Over Temp	#5	4	1	Rise if battery pack temp > 60°C, This value can be set-up
Error Cell Under Temp	#6	5	1	Rise if battery pack temp < 0°C during charge or < -20°C during discharge, These value can be set-up
Error BMS Board over temp	#7	6	1	Rise if BMS temperature > 75°C, This value can be set-up
Error BMS Board under temp	#8	7	1	Rise if BMS temperature < -20°C, This value can be set-up
Error Power Contactor	#19	8	1	Power contactor was opened 3 times within 15 minutes / 3 errors rise within 15 minutes
Lose communication Master	#9	9	1	Slave loses communication with Master
Lose communication Slave	#20	10	1	Master loses communication with Slave
Not used		11	1	
Error current inconsistency	#10	12	1	Slave current isn't consistent with others slaves current
Error voltage bus	#21	13	1	
Error Heater	#22	14	1	
Error Stack	#23	15	1	Error in stack or Error voltage bus stack



Error LifeLine	#24	16	1	LifeLine Error (LifeLine disconnected) 0: No error LifeLine (LifeLine connected or no option LifeLine)
Error Ground	#25	17	1	1 : A grounding default has been detected within the stack
Error IMR	#26	18	1	Input current > IMR and temperature negative
Error IMD	#27	19	1	Output current > IMD and temperature negative
Not used		20	12	Not used
warning Output current	#11	32	1	Rise when output current > 200A, This value can be set-up
warning Input current	#12	33	1	Rise when input current > 105A, This value can be set-up
warning Cell over voltage	#13	34	1	Rise when one cell is > 3680mV, This value can be set-up
warning Cell under voltage	#14	35	1	Rise when one cell is < 3000mV, This value can be set-up
warning Cell Over Temp	#15	36	1	Rise if battery pack temp > 58°C, This value can be set-up
warning Cell Under Temp	#16	37	1	Rise if battery pack temp < 2°C during charge or < -18°C during discharge, These value can be set-up
warning BMS Board over temp	#17	38	1	Rise if BMS temperature > 58°C, This value can be set-up
warning BMS Board under temp	#18	39	1	Rise if BMS temperature < -18°C, This value can be set-up
IMR	#28	40	1	Input current > IMR
IMD	#29	41	1	Output current > IMD
Not used			22bits	Not used



10.2.3.6 Messages Error and Warning for SuperMaster

ID message	Period	Data Length
0x024YY022	500ms when error or warning	8

Note:

- "y" represents the ID of the stack concerned by the messages.

Be careful, for the stack ID, bit 16 in byte 2 is equal to 1 if the stack number is greater than 15.

	Byte 3								Byte 2								Byte 1								Byte 0								
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
	P				Dev				Trans		module				stack				frame		Command ID												
0b	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	y	y	y	y	Y	0	0	0	0	0	0	1	0	0	0	1	0
0x	0				2				4				Y				Y				0		2				2						

message Error/Warning (sent by SuperMaster modules)

Description:

This message is sent by the **SuperMaster** module. It contains error and warning information about the managed stack. This message is sent when there is at least one error or warning. When the event ends, the SuperMaster sends this message twice with the data 0.

Data information:

Data Name	Internal Event Code	First Bit	Length	Comments
Error Output Current	#30	0	1	Output current too high (based on SM Hall sensor measurement) This value can be set-up
Error Input Current	#31	1	1	Input current too high (based on SM Hall sensor measurement) This value can be set-up
Time out Precharge	#32	2	1	1: Time out precharge
Error Power Contactor	#33	3	1	Power contactor is opened 3 times because of error within one hour
Lose communication Main Master	#34	4	1	SuperMaster loses the communication with Main Master
Lose communication SuperMaster	#35	5	1	MainMaster loses the communication with SuperMaster
Lose communication Slave	#36	6	1	SuperMaster loses the communication with slave
Error Voltage bus	#37	7	1	During Precharge : When Bus voltage do not reach 80% of battery voltage When a SuperMaster need to join a pool of stack, If 95% of bus voltage >Stack Voltage>105% of bus voltage
Number Slave Error		8	5	Number of slave modules with an open error status
LifeLine Error	#38	13	1	1: LifeLine Error (LifeLine disconnected) 0: No error LifeLine (LifeLine connected or no option LifeLine)
Ground Error	#39	14	1	1 : A grounding default has been detected within the stack
Error current inconsistency	#40	15	1	Stack current isn't consistent with others stack current measurement
Not Used		16	16	Not used
warning output current	#41	32	1	Output over current warning (based on SM Hall sensor measurement) This value can be set-up
warning input current	#42	33	1	Input over current warning (based on SM Hall sensor measurement) This value can be set-up
Number Slave warning		34	5	Number of slave modules with an open warning status
Not Used		35	25	



10.2.3.7 Status 1 for SuperMaster (and MainMaster)

ID message	Period	Data Length
0x124YY020	1s	8

Note:

- "y" represents the ID of the stack concerned by the messages.

Be careful, for the stack ID, bit 16 in byte 2 is equal to 1 if the stack number is greater than 15.

bit number	Byte 3								Byte 2								Byte 1								Byte 0							
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	P				Dev				Trans				module				stack				frame				Command ID							
0b	0	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	y	y	y	y	y	0	0	0	0	0	0	1	0	0	0	0
0x	1				2				4				Y				Y				0				2				0			

Status 1 message for SuperMaster

Description:

This message is sent by the **SuperMaster**. It contains information about the battery status. The delay between each message is 1s.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
Voltage	0	16	25mV	0V → 1638.375V	Rack voltage calculated by SuperMaster
Current	16	16	100mA	-3276.8 → +3276.7A	Current IN or OUT value, this data is signed. Current IN is positive, Current OUT is negative
SOC	32	16	0.01%	0% → 100.00%	State of Charge in percent
SOH	48	16	0.01%	0% → 100.00%	State of Health in percent



10.2.3.8 Status 2 for SuperMaster (and MainMaster)

ID message	Period	Data Length
0x124YY021	1s	8

Note:

- "y" represents the ID of the stack concerned by the messages.

Be careful, for the stack ID, bit 16 in byte 2 is equal to 1 if the stack number is greater than 15.

	Byte 3								Byte 2								Byte 1								Byte 0								
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
				P				Dev	Trans	module				stack				frame		Command ID													
0b	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	0	y	y	y	y	y	0	0	0	0	0	0	1	0	0	0	0	1
0x	1			2			4			Y			Y			0		2			1												

Status 2 message for SuperMaster

Description:

This message is the same as the Status 1 **SuperMaster** message and contains additional information.

Data information:

Data Name	First Bit	Length	Unit	Range	Comments
Open Power Contactor manually	0	1			Contacteur is manually opened by message 0x30C see chapter 6 : contacteur Open close
Status Power Contactor	1	1			1: Power Contactor closed 0: Power Contactor open
Status Precharge	2	1			1: Precharge active 0: Precharge not active
Number slave in balancing	3	5		0→31	Number of slave in balancing
Number slave inter balancing	8	5		0→31	Number of slave in inter balancing by Super Master
End of charge	13	1			1: end of charge. Battery full
Not used	14	2			
IMR	16	14	100mA	0→1638.3A	Recommended max charge current
IMD	30	12	1A	0→4095A	Recommended max discharge current
Status SuperMaster	42	4			0: Idle 1: Operative 2: Error
Not Used	46	2			
Voltage bus	48	16	100mV	0→6553.6V	Voltage bus measured by SuperMaster



10.2.3.9 Status 1 for Master & Slave

ID message	Period	Data Length
0x12YYY020	1s	8

Note:

- "y1 y2 y3 " represents the transmitter ID, module ID and stack ID concerned by the message.

bit number	Byte 3								Byte 2								Byte 1								Byte 0							
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	P				Dev				Trans				module				stack				frame				Command ID							
0b	0	0	0	1	0	0	1	0	y1	y1	y2	y2	y2	y2	y2	y3	y3	y3	y3	y3	0	0	0	0	0	0	1	0	0	0	0	0
0x	1				2				Y				Y				Y				0				2				0			

Status 1 message pour Master et Slave

Description:

This message is sent by the **Master** and **Slave** modules. The messages provide status information such as voltage, current, SOC, SOH of each module.

Master identifier is 0x12C21020. The priority is low (1), the module number and stack number are 1, the module sender is 0b11 and the frame type is 0x00(TX).

Slave identifier is 0x12yyy020: priority is low (1), module number is to be defined, stack number is 1, module transmitter is 0b10 and frame type is 0b00(TX).

Data information:

Data name	First Bit	Length	Unit	Range	Comments
Voltage	0	16	mV	0→65.535V	Battery pack voltage
Current	16	16	10mA	-327.68→+327.67A	Current IN or OUT value, This data is signed. Current IN is positive, Current OUT is negative
SOC	32	16	0.01%	0% → 100.00%	State of Charge value
SOH	48	16	0.01%	0% → 100.00%	State of Health in percent



10.2.3.10 Status 2 for Master & Slave

ID message	Period	Data Length
0x12YYY021	1s	8

Note:

- "y1 y2 y3" represents the transmitter ID, module ID and stack ID concerned by the message.

	Byte 3								Byte 2								Byte 1								Byte 0								
bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
					P				Dev	Trans		module				stack				frame		Command ID											
0b	0	0	0	1	0	0	1	0	y1	y1	y2	y2	y2	y2	y2	y3	y3	y3	y3	y3	0	0	0	0	0	0	0	1	0	0	0	0	1
0x	1				2				Y				Y				Y				0		2				1						

Status 2 for Master and Slave

Description :

This message is sent by the **Master** and **Slave** modules. It is identical to Status 1 and contains additional information.

Data information:

Data name	First Bit	Length	Unit	Range	Comments
Open Power Contactor manually	0	1			Contactor is manually opened by message 0x30C see chapter 6 : contactor Open close
Status Power Contactor	1	1			1: Power Contactor closed 0: Power Contactor open
Reserved	2	1			
Number cells In Balancing	3	5		0-31 (0→16 cellules)	number of cellules in balancing
Status Balancing Active	8	1			1: At least one slave in balancing 0: no slave in balancing Message available for Master module only
End of charge	9	1			1: end of charge. Battery full
Status heater	10	1			1: heater active
Floating mode	11	1			1: floating mode
Not used	11	4			
IMR	16	16	100mA	0A →6553.5A	Recommended Max charge current Value of IMR [0 - 1000] = 0A - 100A
IMD	32	8	1A	0A → 255A	Recommended Max discharge current Value of IMD: [0 - 150] = 0A - 150A
Temp board	40	8	°C (offset -40)	(0→255) (-40°C →215°C)	BMS board temperature Example: Value temp = 100 => real temp = 100 - 40 = 60°C



Temp pack n°1	48	8	°C (offset -40)	(0→255) (-40°C →215°C)	Temperature of battery bank n°1 Example: Value temp = 100 => real temp = 100 - 40 = 60°C Example: Value = 60 => real temp = 60 - 40 = 20°C
Temp pack n°2	56	8	°C (offset -40)	(0→255) (-40°C →215°C)	Temperature of battery bank n°2 Example: Value = 40 => real temp = 40 - 40 = 0°C



10.2.4 Other CAN bus messages

10.2.4.1 Cell voltage for Master & Slave

ID message	Period	Data Length
0x12yyy103 0x12yyy104 0x12yyy109 0x12yyy110	On request	8

Description :

These messages are sent by **Master** and **Slave**. They contain the cell voltages of the pack (1-16). Each message contains 4 cell voltages.

0x12yyy103: cell voltages 1 to 4
0x12yyy104: cell voltages 5 to 8
0x12yyy109: cell voltages 9 to 12
0x12yyy110: cell voltages 13 to 16

To request a Slave to send these messages, you must use the maintenance message (ID: 0x205) with the Slave ID.

Examples :

Message 0x13021605 (with Cell Voltage Enable = 1) allows Slave 1 of Stack 1 to send the cell voltages (in one go).

0x13041605 (with Cell voltage enable = 1) allows Slave2 stack 1 to send cell voltages (in one go).

Data information:

Data Name	First Bit	Length	Unit	Comments
Cell voltage	0	16	mV	Cell voltage
Cell voltage	16	16	mV	
Cell voltage	32	16	mV	
Cell voltage	48	16	mV	



10.2.4.2 Cell balancing for Master and Slave

ID message	Period	Data Length
0x1Ayyy209	On request	2

Description :

In this message you can see which cells are being balanced.

Data information:

Data Name	First Bit		Length	Comments
Bal. Cell 1	0		1	1 = cell in balancing ; 0 = no balancing
Bal. Cell 2	1		1	
Bal. Cell 3	2		1	
Bal. Cell 4	3		1	
Bal. Cell 5	4		1	
Bal. Cell 6	5		1	
Bal. Cell 7	6		1	
Bal. Cell 8	7		1	
Bal. Cell 9	8		1	
Bal. Cell 10	9		1	
Bal. Cell 11	10		1	
Bal. Cell 12	11		1	
Bal. Cell 13	12		1	
Bal. Cell 14	13		1	
Bal. Cell 15	14		1	
Bal. Cell 16	15		1	



10.2.4.3 Maintenance Message

ID message	Period	Data Length
0x13yyyA05	Request	1

Description :

This message is sent by an **external controller**. The ID of this message is 0x205 with frame type 10: Dreq. This request is used to retrieve information from a Slave.

In SuperMaster/Slave mode, if you want to get the status of the Slaves, you must request "Slave Status" by setting "All Slave status" = 1.

In Master/Slave mode, you don't need to request "Slave Status", because Status 1 and Status 2 are sent periodically by the Master and Slave.

If you want to have the "Slave cell voltage" or/and the "Slave cell balancing" of all Slaves in the same Rack, you have to send this message to each Slave (e.g. message 0x13041605 with 1 data 0x03 to have the "Slave cell voltage" and the "Slave cell balancing" of Slave 2 in Stack 1)

Data information:

Signal Name	First Bit	Length	Comments
Slave cell voltage	0	1	1: Request Slave or SuperMaster to forward cell voltage values
Slave cell balancing	1	1	1: Request Slave SuperMaster to forward cell balancing status
All Slave status	2	1	Request Super Master sent all slaves status1 and 2
Time	3	1	Reserved SuperMaster
Renumber Slave	4	1	Reserved SuperMaster
Reboot all	5	1	Reserved SuperMaster
Slave error/warning	6	1	Request SuperMaster sent slaves status error/Warning message
Slave status 3	7	1	Min/Max Battery voltage/current
Not use	8	4	
Slave status 1	12	1	Request SuperMaster sent one slaves status1
Slave status 2	13	1	Request SuperMaster sent one slaves status2



10.2.4.4 Contactor ON - OFF

ID message	Period	Data Length
0x030yy60C	Request	1

Description :

This message is sent by an **external controller**. It has the ID 0x20C with "Frame type 01: RX".
For example: To control the power contactor of the Master, the message 0x0302160C must be sent.
For the power contactor of Slave2 in stack 1, message 0x304160C must be sent.

This request forces the power contactor to open or close if there is no error.

More details:

The power contactor is permanently controlled by the BMS: it opens in case of a detected error (current error, voltage error...).

As soon as the error is cleared, the contactor closes and reconnects the battery output. In case the power contactor is opened 3 times in 15 minutes, it will remain open for 15 minutes to protect the battery and will be closed automatically afterwards.

Data information:

Data Name	First bit	Length	Comments
Close power contactor	0	1	1 close power contactor
Open power contactor	1	1	1 open power contactor
Close all power contactor	2	1	Reserved
Open all power contactor	3	1	Reserved

10.2.4.5 Heater management

ID message	Period	Data Length
0x03yyyA11	request	4

Description:

This message is sent by an **external controller**. It is used to activate or deactivate the heater.

Data information:

Signal Name	Start Bit	Length	Comments
CtlHeat	0	8	0x01 active heater; 0x00 disable heater
TempHeat	8	8	Target Temperature offset -40 (0°C = 40) Value from 20 to 100 (-20°C to 60°C)
RateHeat	16	8	Rate active heater from 1 to 10
DelayHeat	24	8	delay holding the heater active after target temperature reached (in minute)



10.2.4.6 Parameters

These requests allow managing (reading/writing) parameters for the BMS.

10.2.4.6.1 Message of Parameters

ID message	Period	Data Length
0x12yyy200	On Request	3 or 8

Description:

This message contains the parameters which are saved in E2PROM. You can receive this message by sending the message "Request Parameter". And you can also change the parameters one by one with the message "Set Parameter".

The data length is 3 or 8. The Byte2 of data is the number of parameters in this message. When the byte2 of data is 1, there is one parameter and the data length is 3. When the byte2 is 3, there are three parameters and the data length is 8.

Data information:

Signal Name	First Bit	Length	Comments
Address parameter	0	8	The address of parameter in the E2PROM
Number parameter	8	8	Number of parameters to be sent
Parameter 1	16	16	
Parameter 2	32	16	
Parameter 3	48	16	

10.2.4.6.2 Setting Parameter

ID message	Period	Data Length
0x13yyy600	Request	3

Description:

This message is used to update one parameter.

Data information:

Signal Name	First Bit	Length	Comments
Address parameter	0	8	Address E2PROM you want to write data
Data 1	8	8	The new value of parameter byte 1 write in E2PROM
Data 2	16	8	The new value of parameter byte 2 write in E2PROM

10.2.4.6.3 Request Parameter

ID message	Period	Data Length
0x13yyyA00	Request	0

Description:

This message request BMS to send all parameters. You'll receive several messages 0x2yyy200 with 8 bytes data. Each message has 3 parameters

Data information: No data